Deep Selective Memory Network With Selective Attention and Inter-Aspect Modeling for Aspect Level Sentiment Classification

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***Abstract*—Aspect level sentiment classification aims to recognize the sentiment polarity of each aspect term in a sentence. However, most of the existing methods usually applied the attention mechanism over position-weighted memory and did not consider inter-aspect information. To address these issues, we propose a novel framework for aspect level sentiment classification, Deep Selective Memory Network (DSMN), which selects the context memory dynamically for better guiding the multi-hop attention mechanism and integrates inter-aspect information with deep memory network. By designing a selective attention mechanism based on the distance information between an aspect and its context, DSMN focuses on different parts of the context memory in different memory network layers to capture abundant aspect-aware context information. Besides, to make full use of the inter-aspect information, we also design effective inter-aspect modeling modules to generate both semantic and relation information of the nearby aspects for the desired aspect. We evaluate the advantages of our framework on three benchmark datasets, and experiment results show that our framework achieves state-of-the-art performance.**

***Index Terms*—Aspect level sentiment classification, deep memory network, inter-aspect modeling, selective attention mechanism.**

I. Introduction

ASPECT level sentiment classification aims to identify the sentiment polarity (e.g., positive, negative, or neutral) of each aspect term in a sentence [1]–[3]. Different from sentence-level sentiment analysis, aspect level sentiment classification can gain a more granular understanding of sentiments for each aspect of a given product or service. For example, in the user comment "the price of the product is low, but the quality is poor," the sentiment polarities of the two aspects "price" and "quality" are positive and negative, respectively. It is commonly seen that the same sentence may have different sentiment polarities for different opinion aspects.

Recent years have witnessed a growing interest in applying attention mechanism and memory network [4]–[6] to the task of aspect level sentiment classification. Attention mechanism can focus on important parts of a specific aspect and its contexts [2], [7], [8], while memory network can store context information and apply multi-hop attention [1], [9]. Although these approaches with memory network have achieved promising results, they adopt multi-hop attention on the context memory obtained by the same context in different layers of the memory network, thus lacking the ability to capture abundant context information, namely, simple information like word meanings and complex information like sentence structure. Moreover, to consider the influence of other aspects in the same sentence, attention mechanism or Bidirectional Long Short-Term Memory (BiLSTM) has also been proposed to capture the inter-aspect information [10], [11]. However, these inter-aspect modeling methods consider the information of other aspects in an implicit way.

To solve the above issues, we propose a novel framework of Deep Selective Memory Network (DSMN) for aspect level sentiment classification. In the proposed DSMN, the distance information between the context words and the desired aspect is integrated into the deep memory network, in which the attention mechanism is guided based on the types of dataset (e.g., formal or ungrammatical). Moreover, an inter-aspect modeling component is designed to obtain the semantic information of the nearby aspects in the sentence. To learn the relation between the specific aspect and its nearby aspects, we jointly conduct an inter-aspect relation learning task, which can provide a background for the desired aspect.

Some preliminary results have been published in [12], which is the basic version of the DSMN framework, namely DSMN-I. DSMN-I [12] integrated semantic dependency information into the original context memory to generate dynamic memory for the multi-hop attention network. In addition, a semantic-aware inter-aspect network and a sentence-level context-moment learning network were designed to exploit the semantic information as well as relation information among aspects in the same sentence. Although improved results have been achieved, [12] still has some disadvantages in the task of aspect-based sentiment classification. Firstly, semantic dependency information is not suitable

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for all types of datasets. Specifically, the dependency trees of ungrammatical sentences are usually of low quality and will introduce noises to the model, so semantic information cannot be applied to some datasets like Twitter, which contains more ungrammatical sentences. Secondly, the context moment loss designed in DSMN-I [12] is sentence-level and the same for all aspects in the sentence, which cannot bring more aspect-aware relation information.

To improve the performance of our model, we proposed a general framework of DSMN, including the basic version of DSMN-I and an advanced version, namely DSMN-II. In the proposed DSMN-II, different kinds of distance measures are used to generate dynamic memory based on the type of data, and aspect-level relation information between the desired aspect and its nearby aspects is learned instead of sentence-level relation information. Comparing with our previous work [12], this version includes the following new contributions.

- Proposing two strategies to generate the dynamic memory for different hops of memory network according to the type of data;
- Further proposing two ways to select the semantic-based distance and the location-based distance, and showing automatic weighting selection is better than the simple averaging selection.
- Designing an aspect-level moment learning task instead of sentence-level moment learning task [12] to better introduce the relation information between the specific aspect and its nearby aspects;
- Improving results on the three benchmark datasets compared to [12].

II. Related Work

In the task of aspect level sentiment classification, the aspects in the sentence have already been extracted. To give a complete overview of aspect level sentiment classification, in this section we will briefly review the aspect extraction firstly. We will then present some classic models that have been applied to the aspect level sentimental classification, and the closely-related methods based on attention and memory network, with promising results reported.

A. Aspect Extraction

Traditionally, some rule-based methods [13], which exploit hand-crafted rules about some syntactic relationships, were proposed to solve the problem of aspect extraction. Recently, many works have concentrated on leveraging deep neural networks to tackle the task of aspect extraction, e.g., using CNNs [14], [15], RNNs [16]–[18], multi-task neural networks [19], [20] and so on. For instance, [15] proposed a CNN model employing two types of pre-trained embeddings for aspect extraction, namely general-purpose embeddings and domain-specific embeddings; [17] designed two LSTMs equipped with extended memories and neural memory operations to jointly handle the extraction tasks of aspects and opinions via memory interactions.

B. Classic Aspect Level Sentiment Classification

Hand-crafted features are used for aspect level sentiment classification with supervised learning approaches [21]–[28]. [21] applied some rules based on part-of-speech tagging and dependency parsing to identify the sentiment polarity of the specific aspect. [24] incorporated syntactic features to distinguish the texts used for expressing sentiments towards different aspects. [26] extracted some n-gram features, lexicon features, and parsing features to predict the polarities of aspects. However, these models fail to model the semantic relation between an aspect and its context, which is critical for aspect level sentiment analysis.

Classic deep learning models, such as Convolutional Neural Networks (CNNs), Recursive Neural Networks (RNNs) and Long Short-Term Memory (LSTM), have been used to extract various features from aspects and contexts automatically. [29] integrated CNNs with gating mechanisms to design a simple and easily-parallelized model for aspect level sentiment classification; [3] designed a transformation network (TNet), which adopted CNNs to generate context features and then proposed a component to generate aspect-aware representations; To learn aspect-specific semantic features, [30] used an aspect-aware RNN network over the dependency tree of the sentence. [31] employed two Target-Dependent LSTM (TD-LSTM) networks to separately learn the left and right context of the desired aspect and adopted a feed-forward network to classify the sentiment of the aspect. However, the interaction between aspects and their contexts is not well exploited in these methods, which are incapable of explicitly exhibiting important context clues of an aspect.

C. Methods Based on Attention and Memory Network

Attention mechanism has been widely used in sentiment classification. [7] was the first to exploit attention mechanism to aspect level sentiment classification by designing the attention-based LSTM with aspect embedding (ATAE-LSTM), which simply concatenates the aspect representation into the sentence representation. In order to exploit the information of context, [32] proposed a bidirectional LSTM with attention and gates (BILSTM-ATT-G) to measure the importance of context; [8] designed an interactive attention network (IAN) to make the representations of aspects and contexts to interact with each other. In addition, [33] promoted the LSTM network with attention mechanism and common-sense knowledge; and [34] enabled rich dyadic interaction between aspect and document.

Memory network is also used to capture the context information. [9] was the first to apply deep memory network (MemNet), in which multiple attention layers are applied over an external memory and location attention is integrated into content location. Moreover, some deep learning architectures like CNN/LSTM/GRU are integrated with memory network to better solve the task of aspect level sentiment classification. For instance, [1] proposed a recurrent attention network, which used a BiLSTM model to encode the context and combined all attention outputs in a nonlinear way with a Gated Recurrent

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Unit (GRU) model; [35] proposed a convolutional memory network (Conv-Memnet) with an attention mechanism; [2] applied sentence-level content attention mechanism and then use two GRU networks to learn the weights of the memory.

Inter-aspect relation has also been integrated into aspect level sentiment classification, and some effective methods based on inter-aspect modeling have been proposed. The attention mechanism has been used in the method [10] to model the relationships among aspects while a BiLSTM network is applied in the method [11] to exploit the information of the nearby aspects.

Although the attention mechanism, memory network, and existing inter-aspect modelings have been introduced into previous methods, which have achieved promising results on aspect level sentiment classification, there are still some problems unsolved.

Firstly, the context memory generated [1], [9] is the same in the different layers of the memory network, thus weakening the ability to capture different context information on different computational layers, namely, simple information like word meanings and complex information like sentence structure.

Secondly, the aspect level information has been well studied, however, the global information to describe the inter-aspect discrimination of the sentence, including the sentence level sentiment and the relation among aspects, is not well exploited in previous methods [1], [9]–[11]. For example, given the sentence "the service and the quality of your product are good.," it is easy to know the sentiment results of "service" and "quality" if we know the sentiment of the sentence is positive and the relation between the two aspects is coordination.

III. Deep Selective Memory Network

To address the issues above, we proposed a novel framework for aspect level sentiment classification, Deep Selective Memory Network (DSMN). In the proposed framework, the distance information between the context and the desired aspect is used to guide the multi-hop attention mechanism in the memory network. In addition, two inter-aspect modeling modules are designed to learn the inter-aspect knowledge, including the relation information between the specific aspect and its nearby aspects as well as the semantic information of nearby aspects. Finally, a fusion component is designed to combine the features learned by the above modules. Based on the proposed framework, we design two models, namely, the basic model DSMN-I and the advanced one DSMN-II. DSMN-I exploits semantic-based information as well as sentence-level inter-aspect relation information to better get the sentiment polarities of the aspects, while DSMN-II employs different types of distances as well as the aspect-level relation learning to enhance the basic model. Since the implementation details of DSMN-I has been well illustrated in [12], we will focus on the design of DSMN-II and illustrate how it improves DSMN-I in the following description of DSMN.

A. Overview of DSMN

Same as previous methods [3], [35], the goal of our model is to identify the sentiment of the extracted aspects. To avoid aspect information from interfering with context modeling, we replace each aspect, which may be a word or a phrase, with a placeholder.

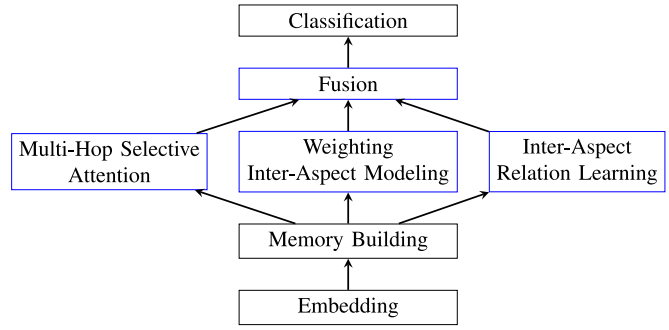

Fig. 1. The overall architecture of DSMN. The modules with blue-colored border are the proposed modules of DSMN.

The architecture of DSMN is shown in Fig. 1, which is composed of seven modules: embedding module, memory building module, multi-hop selective attention module, weighting inter-aspect modeling module, inter-aspect relation learning module, fusion module, and classification module. Among these modules, the multi-hop selective attention module, the weighting inter-aspect modeling module, the inter-aspect relation learning module, and the fusion module are newly proposed in the DSMN framework.

In the embedding module, we use a word embedding lookup table to convert most words to their vectors and initialize other out-of-vocabulary (OOV) words by randomly sampling from a uniform distribution. Here unsupervised methods like Word2Vec [36], Glove [37] or FastText [38], [39] can be used to learn the table.

To generate an external memory, a BiLSTM network, which can capture the relationship between words in a sentence, is adopted in the memory building module. After modeling, the originally-generated context memory is

$$\boldsymbol{M}_c = \{\boldsymbol{m}_1^*, \boldsymbol{m}_2^*, \ldots, \boldsymbol{m}_n^*\} \tag{1}$$

where n stands for the length of sentence and $\boldsymbol{m}_i^* = (\overrightarrow{\boldsymbol{h}_i^*}, \overleftarrow{\boldsymbol{h}_i^*})$, with $\overrightarrow{\boldsymbol{h}_i^*}$ and $\overleftarrow{\boldsymbol{h}_i^*}$ denoting the hidden states generated by the forward and the backward LSTMs of BiLSTM, respectively.

The multi-hop selective attention module selects different slices of memory dynamically for different computational layers based on semantic information, location information, or both of them, and then applies the attention mechanism to generate the aspect-aware feature $\boldsymbol{v}_{sa}$ for the desired aspect.

To recognize the sentiment of an aspect with the help of other aspects in the same sentence, the weighting inter-aspect modeling module and the inter-aspect relation learning module are designed. The former module uses the distance information between the specified aspect and other aspects to get the semantic information of other aspects, while the latter module applies an inter-aspect relation learning task to learn the relation between the desired aspect and its nearby aspects. The outputs of the two modules are denoted as $\boldsymbol{v}_{im}$ and $\boldsymbol{v}_{ir}$.

To control how much inter-aspect information the final classification layer will exploit, we simply design a fusion layer, which will be detailed in Part E of Sec. III. And then we can get the final feature for classification:

$$\boldsymbol{v} = fuse(\boldsymbol{v}_{sa}, \boldsymbol{v}_{im}, \boldsymbol{v}_{ir}) \tag{2}$$

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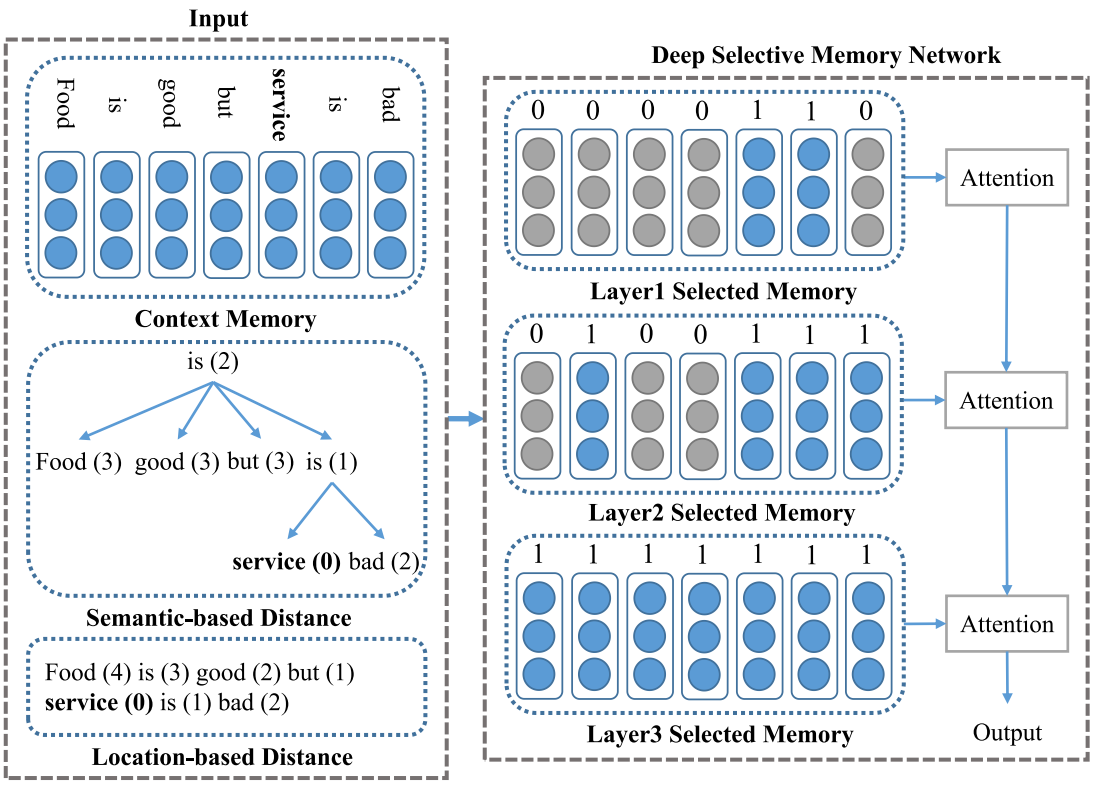

Fig. 2. Architecture of multi-hop selective attention. The left part of the figure illustrates the input of the deep selective memory network, namely, the context memory $\{\boldsymbol{m}_1^*, \ldots, \boldsymbol{m}_i^*, \ldots, \boldsymbol{m}_n^*\}$, the semantic-based distance shown in the dependency parsing tree, and the location-based distance shown in the input sequence. The word nodes are in the form of $word(dist(w_i, w_t))$, where $dist(w_i, w_t)$ is the distance from the aspect w_t to the context word w_i. The right part shows how the deep selective memory network works. Here gives an example that applies semantic-based selection, and the attention mechanism is only applied on the blue-colored memory slices based on Equation (5).

where $\boldsymbol{v}_{sa}$, $\boldsymbol{v}_{im}$ and $\boldsymbol{v}_{ir}$ are the outputs of multi-hop selective attention module, weighting inter-aspect modeling module and inter-aspect relation learning module, respectively.

The $softmax$ function is used in the classification module, which adopts a feed-forward network to project $\boldsymbol{v}$ into the space of the targeted C classes.

$$\boldsymbol{y} = softmax(\boldsymbol{W}_c \boldsymbol{v} + \boldsymbol{b}_c) \tag{3}$$

where $\boldsymbol{y} = [y_1, \ldots, y_i, \ldots, y_C] \in R^C$ is the predicted probability, and $\boldsymbol{W}_c$ and $\boldsymbol{b}_c$ are the weight matrix and bias, respectively. The model is finally trained by minimizing the sum of the cross entropy, the inter-aspect relation learning loss and the L_2 regularization term:

$$L = -\sum_{t \in T} \left(\sum_{i=1}^{C} g_i^t log y_i^t + \lambda_m l_m^t \right) + \lambda \|\theta\|_2^2 \tag{4}$$

where T stands for the aspect set, $\boldsymbol{y}^t = [y_1^t, \ldots, y_i^t, \ldots, y_C^t] \in R^C$ is the estimated probability for aspect t, and $\boldsymbol{g}^t = [g_1^t, \ldots, g_i^t, \ldots, g_C^t] \in R^C$ is the ground truth for aspect t, which is represented by a one-hot vector where the element for the correct polarity is 1. Here λ is the weight of L_2 regularization term and λ_m is the weight of the inter-aspect relation learning loss. The term of l_m^t is the inter-aspect relation learning loss for the aspect t, which will be detailed in the following section.

B. Multi-Hop Selective Attention

In this section, we present the multi-hop selective attention module, as shown in Fig. 2. The module firstly generates dynamic selective memory for the multi-hop memory network based on semantic information, location information, or both of them, and then applies attention mechanism between the desired aspect and the selected context memory at every computational hop. After applying recurrent attention, we can take the output from the final layer of the multi-hop attention network $\boldsymbol{o}_l$ as the output $\boldsymbol{v}_{sa}$.

1) Dynamic Selective Memory: Since the originally-generated memories are the same for different aspects in the same sentence, it's necessary to generate aspect-specific memory for different aspects. To address the issue, [1], [9] weighted the original memory according to the absolute distance from a context to the specific aspect, and [40] proposed position embedding to utilize the location information of context. However, these models only consider the location information while ignoring semantic information, which may be helpful for some domains containing grammatical sentences mostly. Moreover, these methods generate aspect-aware memory with the same context in the multi-hop memory network. Therefore, we design a dynamic selective memory with location information or semantic information, which can not only adapt different types of data (e.g., formal or

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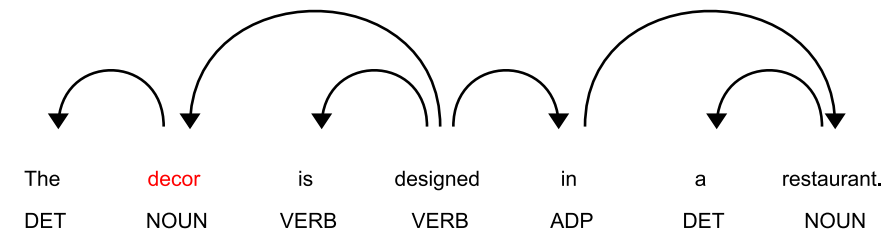
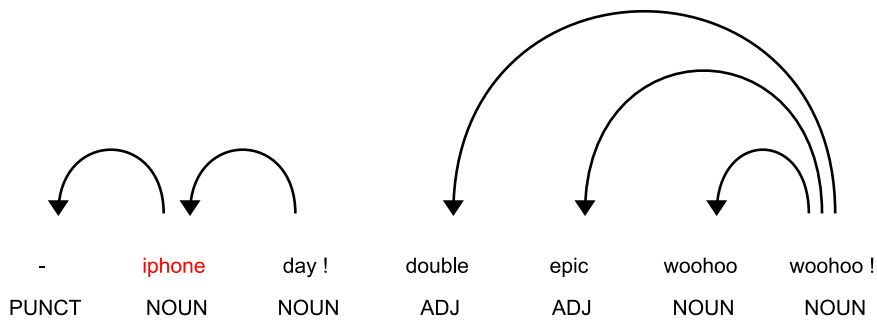


(a) A formal sentence example.



(b) An ungrammatical sentence example.

Fig. 3. Comparison between examples of different dataset types. As for the formal sentence example, the semantic information is accurate so the sentiment of the aspect "decor," namely neutral, can be predicted by semantic information. As for the ungrammatical sentence example, the semantic information is not accurate enough, so the context word "woohoo" for predicting the sentiment of the aspect "iphone" can't be selected based on semantic information.

ungrammatical), but also provide rich context information (e.g., simple information like word meanings and complex information like sentence structure) for the memory network.

In the basic model DSMN-I, semantic information is exploited to select memory dynamically. However, it may obtain inaccurate semantic information for some datasets with lots of ungrammatical sentences. Therefore, we proposed the advanced model DSMN-II, which considers both semantic and location information to adapt the model to different types of datasets.

We can utilize semantic information or location information to encode the relation between the specific aspect and its context words. The two types of information are suitable for different types of data. As Fig. 3 is shown, the datasets like product reviews, which relatively contain more formal sentences, can get accurate dependency information, so semantic information is suitable for these datasets. On the contrary, the datasets like tweets, which contain more ungrammatical sentences, may get the wrong dependency information, so location information will be more suitable to guide the model. To adapt different types of data, we design two ways to compute $dist(w_i, w_t)$, namely the distance from the aspect word w_t to the context word w_i, which is used later.

- **Semantic-based Distance**: $dist(w_i, w_t)$ is the length of the path from the aspect word w_t to the context word w_i in the semantic dependency tree of the sentence, which is obtained by a open-source library spaCy;[1]
- **Location-based Distance**: $dist(w_i, w_t)$ is the absolute distance from the aspect word w_t to the context word w_i in the original sentence sequence.

[1][Online]. Available: https://spacy.io/

Based on the two types of distance, we can obtain the selection information for each layer of the memory network. Specifically, at the lower layers, the closer context words will be selected. As the memory network gets deeper, the context words which are more distant to the desired aspect will be considered.

$$sel_i^l = \begin{cases} 1, & \text{if } dist(w_i, w_t) \leq l \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

For the semantic-based distance, the corresponding selection information is denoted as $sel_i^l|_{sem}$, and for the location-based distance, the selection information is $sel_i^l|_{loc}$.

For different layers in the deep memory network, the model selects different parts of memory depending on the distance between each context word and the aspect. To further the effectiveness of the two types of distance, we design four strategies to exploit the obtained distances for memory selection.

The first two strategies are **semantic-based selection** and **location-based selection**, respectively. At l-th layer, each memory unit is selected with the distance information can be represented as follows:

$$\boldsymbol{m}_i^l = sel_i^l * \boldsymbol{m}_i^* \tag{6}$$

where $\boldsymbol{m}_i^*$ stands for the original memory unit generated by the BiLSTM. sel_i^l is $sel_i^l|_{sem}$ for semantic-based selection or $sel_i^l|_{loc}$ for location-based selection.

As for semantic-based selection or location-based selection, only one type of distance is exploited. However, exploiting both of the two types of distance may provide more information to better select the context memory. Therefore, we design two strategies, namely the **averaging selection** and the **weighting**

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selection, to combine the information of both semantic-based selection and location-based selection.

The averaging selection treats semantic-based information and location-based information equally, and simply computes the mean value of the two types of selection information as the final selection information.

$$\boldsymbol{m}_i^l = \frac{sel_i^l|_{sem} + sel_i^l|_{loc}}{2} * \boldsymbol{m}_i^* \tag{7}$$

Unlike the averaging selection, the weighting selection computes the weights of the two distances for each sample.

$$\boldsymbol{m}_i^l = (p_{sel} * sel_i^l|_{sem} + (1 - p_{sel}) * sel_i^l|_{loc}) * \boldsymbol{m}_i^* \tag{8}$$

where $p_{sel} \in [0, 1]$ is the learned probability of exploiting semantic-based information. To learn the probability p_{sel}, we first apply attention mechanism over the context memory to better identify whether the sentence is grammatical or not.

$$\gamma_i = softmax(\boldsymbol{W}_\gamma \boldsymbol{m}_i^* + \boldsymbol{b}_\gamma) \tag{9}$$

where $\boldsymbol{W}_\gamma$ and $\boldsymbol{b}_\gamma$ are weight matrices and biases. And then we can predict the probability of using semantic information.

$$p_{sel} = sigmoid\left(\boldsymbol{W}_{sel}\left(\sum_i^n \gamma_i * \boldsymbol{m}_i^*\right) + \boldsymbol{b}_{ir}\right)) \tag{10}$$

With the above design, the final selected memory generated at l-th layer is $\{\boldsymbol{m}_1^l, \boldsymbol{m}_2^l, \ldots, \boldsymbol{m}_n^l\}$.

2) Aspect Modeling: Since an aspect may be a phrase, it's necessary to encode the aspect in an effective way, rather than simply taking an average of its constituting word vectors [1], [9]. To obtain the aspect representation $\boldsymbol{v}_a$, we use a BiLSTM layer and a self-attention layer.

Similar to encoding context, the aspect's hidden states obtained by employing BiLSTM is $\boldsymbol{H} = \{\boldsymbol{h}_1, \boldsymbol{h}_2, \ldots, \boldsymbol{h}_{n_a}\}$, where n_a is the length of the aspect. To capture the important part of the aspect, we apply an attention layer:

$$s_i = softmax(\boldsymbol{W}_1 tanh(\boldsymbol{W}_2 \boldsymbol{h}_i)) \tag{11}$$

where W_1 and W_2 are weight matrices. Finally we can get the representation of aspect $\boldsymbol{v}_a$:

$$\boldsymbol{v}_a = \sum_{i=1}^{n_a} s_i \boldsymbol{h}_i \tag{12}$$

3) Attention Mechanism on Memory: With the selected memory and the obtained aspect representation, we can use the aspect representation $\boldsymbol{v}_a$ to get the normalized attention score α_i^l as follows:

$$\alpha_i^l = softmax(\boldsymbol{m}_i^l \boldsymbol{W}_\alpha \boldsymbol{v}_a)) \tag{13}$$

where $\boldsymbol{W}_\alpha$ is a trainable weight matrix, $\boldsymbol{m}_i^l$ and $\boldsymbol{v}_a$ stand for the memory unit and the aspect representation, respectively. And α_i^l is the normalized attention score of each context memory unit $\boldsymbol{m}_i^l$ at l-th layer of the memory network.

Then the attention output $\boldsymbol{v}_l$ is calculated as:

$$\boldsymbol{v}_l = \sum_{i=1}^{n} \alpha_i^l \boldsymbol{m}_i^l \tag{14}$$

4) Multi-Hop Attention Combination: In the multi-hop memory network, how to combine the attention results from different computational hops is an important problem. Inspired by [3], there are two approaches to combine the results: lossless forward and gated forward.

Lossless forward treats every attention layer equally and combines the attention results by simply summing up them.

$$\boldsymbol{o}_l = \boldsymbol{v}_l + \boldsymbol{o}_{l-1} \tag{15}$$

where $\boldsymbol{o}_0$ is a vector of 0's.

Gated forward employs a gating function to control the passed proportions of the attention features and the output of the last layer [41]:

$$\boldsymbol{o}_l = \boldsymbol{v}_l \odot \boldsymbol{T} + \boldsymbol{o}_{l-1} \odot (1 - \boldsymbol{T}) \tag{16}$$

where the gate $\boldsymbol{T}$ is designed to control how much of the outputs from the previous layer and the current layer used. The formula of $\boldsymbol{T}$ is defined as follows:

$$\boldsymbol{T} = \sigma(\boldsymbol{W}_t \boldsymbol{o}_{l-1} + \boldsymbol{b}_t) \tag{17}$$

where σ stands for the $sigmoid$ activation function, and $\boldsymbol{W}_t$ and $\boldsymbol{b}_t$ are weight matrices and biases, respectively.

Since gated froward can better control how much output of the current layer exploited for the classification of the sentiment polarities, we apply gated forward to combine the attention results from all computational layers.

C. Weighting Inter-Aspect Modeling

As for a sentence, there may be one or more aspects. Intuitively, the information of other aspects in the same sentence is useful for the sentiment classification of the desired aspect. Therefore, we design a weighting inter-aspect modeling to obtain the feature of other aspects in the same sentence.

After applying the multi-hop selective attention, we can get $p - 1$ aspect-aware representations for all p aspects in the sentence except the current aspect. Then we stack the representations as the aspect memory according to the occurrence order of their corresponding aspect-terms. The aspect memory can be represented as follows:

$$\boldsymbol{M}_{sa} = \{\boldsymbol{v}_{sa_1}, \boldsymbol{v}_{sa_2}, \ldots, \boldsymbol{v}_{sa_{p-1}}\} \tag{18}$$

Intuitively, the closer an aspect in the same sentence to the desired aspect, the more important it is. Therefore, we weight the memory according to the distance between a context word and the aspect:

$$\boldsymbol{M}'_{sa} = \{\boldsymbol{v}'_{sa_1}, \boldsymbol{v}'_{sa_2}, \ldots, \boldsymbol{v}'_{sa_{p-1}}\} \tag{19}$$

where $\boldsymbol{v}'_{sa_i} = \lambda_i * \boldsymbol{v}_{sa_i}$ and λ_i is the distance weight, which is calculated as follows:

$$\lambda_i = 1 - \frac{dist(w_i, w_t)}{d} \tag{20}$$

where $dist(w_i, w_t)$ is the distance between the current aspect w_i and the desired aspect w_t. Besides, d stands for the depth of the dependency tree if choosing semantic-based distance or the length of sentence sequence if applying location-based distance.

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For DSMN-II, it adopts the same strategy as the multi-hop selective attention module to compute λ_i.

To get the final representation $\boldsymbol{v}_{im}$ with a cheaper cost, the advanced model DSMN-II simply sums up all the weighted slices of the aspect memory.

$$\boldsymbol{v}_{im} = \sum_{i}^{p-1} \boldsymbol{v}'_{sa_i} \tag{21}$$

Compared to applying the self attention mechanism, summing up is cheaper and more reasonable since it directly considers the distances between the specific aspect and other aspects as the importance weights of the memory slices.

D. Inter-Aspect Relation Learning

Some sentiment relations among aspects, like comparisons and coordinations, are commonly existed in a sentence with multiple aspects. If the relation between the desired aspect and its nearby aspects is introduced to the model, we can identify whether the final representation of the nearby aspects v_{im} is useful and how they assist the sentiment classification for the desired aspect. In this section, we propose an inter-aspect relation learning task to capture the relation between the desired aspect and its nearby aspects.

Denote -1, 0 and 1 the negative, neutral and positive sentiment polarities, respectively, we can obtain all the labels of the aspects $\{lab_1, lab_2, \ldots, lab_p\}$, where $lab_i \in \{-1, 0, 1\}$. Then we can compute the sentence-aware and aspect-aware context moments to portray the sentiment and relation information of the other aspects in the same sentence:

$$\mu_1 = \frac{1}{p}\sum_{i=1}^{p} lab_i \tag{22}$$

$$\sigma_1^2 = \frac{1}{p}\sum_{i=1}^{p} (lab_i - \mu_1)^2 \tag{23}$$

$$\mu_2 = \frac{1}{p}\sum_{i=1}^{p} \lambda_i * lab_i \tag{24}$$

$$\sigma_2^2 = \frac{1}{p}\sum_{i=1}^{p} \lambda_i * (lab_i - \mu_2)^2 \tag{25}$$

where μ_1 and σ_1^2 are the sentence-aware mean value and variance of the sentiment distribution for the sentence, while μ_2 and σ_2^2 are the aspect-aware mean value and variance of the sentiment distribution for the desired aspect. To pay more attention to the desired aspect and its nearby aspects, DSMN-II employs the context moment learning task with the aspect-level context moment instead of sentence-level one in DSMN-I. Moreover, since the sentiment information of nearby aspects is learned by the weighting inter-aspect modeling, we adopt aspect-aware variance to portray the relationship between the desired aspect and its nearby aspects.

To learn the estimation of the aspect-aware variance, the module firstly converts the original context memory to the aspect-aware weighted memory:

$$\boldsymbol{M}'_c = \{\boldsymbol{m}'_1, \boldsymbol{m}'_2, \ldots, \boldsymbol{m}'_n\} \tag{26}$$

where $\boldsymbol{m}'_i = \lambda_i * \boldsymbol{m}^*_i$ and λ_i is the weight calculated based on the distance between the current context word w_i and the desired aspect w_t.

To capture the important part for estimating the aspect-aware variance value, we apply an attention layer over the weighted context memory.

$$\beta_i = softmax(\boldsymbol{W}_\beta \boldsymbol{m}'_i + \boldsymbol{b}_\beta) \tag{27}$$

where $\boldsymbol{W}_\beta$ and $\boldsymbol{b}_\beta$ are weight matrices and biases. And then we can get the inter-aspect relation representation for estimating the inter-aspect variance value.

$$\boldsymbol{v}_{ir} = \boldsymbol{W}_{ir}\left(\sum_{i}^{n} \beta_i * \boldsymbol{m}'_i\right) + \boldsymbol{b}_{ir}) \tag{28}$$

Finally we adopt a feed-forward network to get the estimation value.

$$\hat{\sigma}^2 = \sigma(\boldsymbol{W}_{var}(\boldsymbol{v}_{ir}; \boldsymbol{v}_{im}) + \boldsymbol{b}_{var})) \tag{29}$$

To train the inter-aspect relation learning module, an aspect-aware variance learning loss for the desired aspect t is defined.

$$l_m^t = \begin{cases} \sigma^2 ln(\hat{\sigma}^2) + (1-\sigma^2) ln(1-\hat{\sigma}^2), & \text{if } p \neq 1 \\ 0, & \text{if } p = 1 \end{cases} \tag{30}$$

It's obvious that the aspect-aware variance learning loss should be ignored when there is only one aspect in the sentence.

E. Fusion Module

To combine the features learned by the above modules, DSMN-I employs the concatenation operation to get the final feature. However, it cannot control how much inter-aspect information the final classification layer will exploit. Therefore, DSMN-II designs an advanced fusion module, as shown in Fig. 4. With the knowledge of the context information of the current aspect $\boldsymbol{v}_{sa}$ and the relation between it and its nearby aspects $\boldsymbol{v}_{ir}$, we can design a fusion gate to control how much inter-aspect global information the fusion module uses.

$$\boldsymbol{T}_f = \sigma(\boldsymbol{W}_{glo}(\boldsymbol{v}_{sa}; \boldsymbol{v}_{ir}) + \boldsymbol{b}_{glo}) \tag{31}$$

where $\boldsymbol{W}_{glo}$ and $\boldsymbol{b}_{glo}$ are trainable matrix and bias, respectively. And then we can use the gate result to obtain the final representation:

$$\boldsymbol{v} = \boldsymbol{v}_{sa} \odot \boldsymbol{T}_f + \boldsymbol{v}_{glo} \odot (1 - \boldsymbol{T}_f) \tag{32}$$

where $\boldsymbol{v}_{glo}$ is the global representation generated based on the feature of the nearby aspects $\boldsymbol{v}_{im}$ and the relation information $\boldsymbol{v}_{ir}$.

$$\boldsymbol{v}_{glo} = \boldsymbol{W}_{glo}(\boldsymbol{v}_{ir}; \boldsymbol{v}_{im}) + \boldsymbol{b}_{glo} \tag{33}$$

IV. Experiments

A. Experiment Setup

To verify the effectiveness of DSMN, we conduct experiments on three benchmark datasets: Laptop and Restaurant reviews

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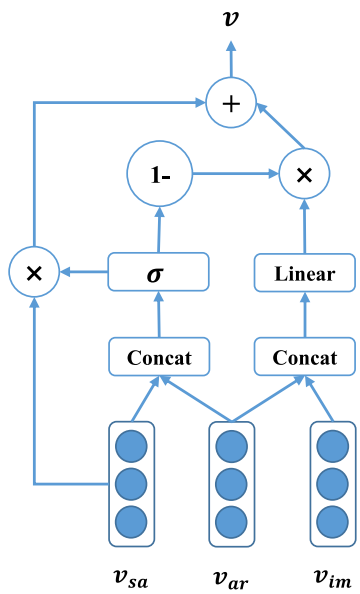

Fig. 4. Architecture of fusion module.

TABLE I
DETAILS OF THE EXPERIMENTAL DATASETS

Dataset	Positive		Negative		Neutral	
	train	test	train	test	train	test
Laptop	994	341	870	128	464	169
Restaurant	2164	728	807	196	637	196
Twitter	1567	174	1563	174	3127	346

from SemEval 2014 Task 4 [42] and Twitter built by [30]. The details of the experimental datasets are shown in Table I. The reviews of Laptop and Restaurant are labeled with four sentiment polarities: positive, neutral, negative, and conflict. Same to previous methods [1], [3], we remove the conflict category as the number of conflict samples is very small and make the dataset extremely unbalanced. In addition, the sentences in the Twitter dataset all contain only one aspect, so our contribution to inter-aspect modeling does not be applied in this dataset. Since both Laptop and Restaurant datasets contain more formal sentences, we employ the semantic-based selection on these two datasets. Conversely, we apply the location-based selection to the Twitter dataset as it contains more ungrammatical sentences. Further discussion on different selection strategies is detailed in Part B of Sec. V.

The embedding lookup table used in experiments is pre-trained by GloVe [37]. The out-of-vocabulary (OOV) words are initialized by sampling from the uniform distribution $U(-0.25,\ 0.25)$. Moreover, the dimension of word vectors is 300, and the dimension of LSTM hidden states and output representation is 50.

We employ Adam Optimizer [43] to train our model and the hyper-parameters are shown in Table II. In addition, we employ early stopping to ease over-fitting.

Evaluation metrics of the experiments are Accuracy and Macro-Averaged F1. The Macro-Averaged F1 is computed by first computing the F1-score of every class (positive, negative, or neutral) and then computing the average value. Therefore, compared to Accuracy, Macro-Averaged F1 is more appropriate for datasets with unbalanced classes.

TABLE II
SETTING OF HYPER-PARAMETERS

Hyper-parameters	Laptop	Restaurant	Twitter
batch size	25	16	64
learning rate	0.002	0.002	0.002
λ	0.001	0.001	0.001
λ_m	10.0	10.0	0.0
dropout rate	0.5	0.5	0.5

TABLE III
EXPERIMENTAL RESULTS OF COMPARISON WITH RECENT METHODS ON LAPTOP (%)

Models	ACC	Macro-F1
TD-LSTM	71.83	68.43
MemNet	70.33	64.09
BILSTM-ATT-G	74.37	69.90
RAM	75.01	70.51
Conv-Memnet	76.37	72.10
CNN-ASP	72.46	65.31
TNet	76.54	71.75
DSMN-I	**77.59**	73.61
DSMN-II	77.27	**73.91**

B. Compared Methods

To evaluate the performance of DSMN, we compare our models to some recent state-of-the-art methods:

- TD-LSTM [31]: It employs two LSTM networks to encode the preceding and following context of the desired aspect separately, then performs predictions with concatenated context features;
- MemNet [9]: It applies memory network over the word embeddings and predicts sentiments based on the output of the final layer;
- BILSTM-ATT-G [32]: It models contexts with attention-based LSTMs and introduces gates to measure the importance of context for the prediction;
- RAM [1]: It adopts a multi-hop network where each layer consists of an attention layer and a GRU cell to learn the sentence representation;
- Conv-Memnet [35]: It captures both words and multi-word expressions to better learn the aspect-aware sentence representation;
- CNN-ASP [3]: It is a CNN-based model that directly concatenates aspect representation to each word embedding;
- TNet [3]: It employs CNNs to extract salient features and a component to generate aspect-specific representations.

C. Performance on Laptop Dataset

We firstly conduct experiments on the Laptop dataset and the results of performance comparison are shown in Table III. From the experimental results, we can observe that DSMN, including DSMN-I and DSMN-II, outperforms all the competing methods significantly. Compared to DSMN-I, DSMN-II achieves better performance as it gets a 0.3% higher Macro-F1, which can better reflect the effectiveness of the model than the Accuracy (ACC) in multi-class classification problems. With improved inter-aspect

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TABLE IV
CASE STUDY ON LAPTOP

Examples	TNet	DMMN-I	DSMN-II
1. Great **[price]**$_+$ free **[shipping]**$_+$ what else can I ask for ! !	[price]$_+$ (✓), [shipping]$_+$ (✓)	[price]$_+$ (✓), [shipping]$_+$ (✓)	[price]$_+$ (✓), [shipping]$_+$ (✓)
2. My laptop with **[Windows 7]**$_-$ crashed and I did not want **[Windows 8]**$_-$.	[Windows 7]$_0$ (✗), [Windows 8]$_0$ (✗)	[Windows 7]$_-$ (✓), [Windows 8]$_-$ (✓)	[Windows 7]$_-$ (✓), [Windows 8]$_-$ (✓)
3. On **[start up]**$_-$ it asks endless questions just so **[itune]**$_-$ can sell you more of their products.	[start up]$_+$ (✗), [itune]$_+$ (✗)	[start up]$_+$ (✗), [itune]$_+$ (✗)	[start up]$_-$ (✓), [itune]$_-$ (✓)
4. The first time I used the **[card reader]**$_-$ it took half an hour and a pair of tweezers to **[the card]**$_-$.	[card reader]$_0$ (✗), [the card]$_0$ (✗)	[card reader]$_0$ (✗), [the card]$_0$ (✗)	[card reader]$_0$ (✗), [the card]$_0$ (✗)
5. A very important feature is **[Firewire 800]**$_+$ which in my experience works better then **[USB3]**$_-$ (in PC enabled with **[USB3]**$_0$) I was not originally sold on the **[Mac OS]**$_-$ I felt it was inferior in many ways To **[Window 7]**$_+$.	[Firewire 800]$_+$ (✓), [USB3]$_+$ (✗), [USB3]$_-$ (✗), [Mac OS]$_-$ (✓), [Window 7]$_-$ (✗)	[Firewire 800]$_+$ (✓), [USB3]$_-$ (✓), [USB3]$_0$ (✓), [Mac OS]$_0$ (✗), [Window 7]$_0$ (✗)	[Firewire 800]$_+$ (✓), [USB3]$_-$ (✓), [USB3]$_0$ (✓), [Mac OS]$_-$ (✓), [Window 7]$_-$ (✗)
6. It's been time for a new laptop, and the only debate was which **[size]**$_0$ of the Mac laptops, and whether to spring for the **[retina display]**$_0$.	[size]$_+$ (✗), [retina display]$_0$ (✓)	[size]$_0$ (✓), [retina display]$_0$ (✓)	[size]$_+$ (✗), [retina display]$_0$ (✓)

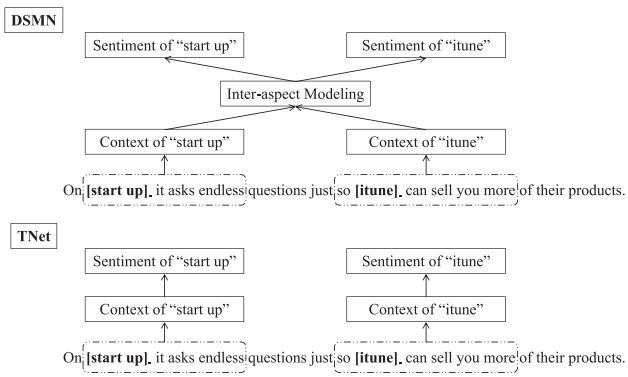

Fig. 5. Comparison between DSMN and TNet.

TABLE V
EXPERIMENTAL RESULTS OF COMPARISON WITH BASELINES ON RESTAURANT (%)

Models	ACC	Macro-F1
TD-LSTM	78.00	66.73
MemNet	78.16	65.83
BILSTM-ATT-G	80.38	70.78
RAM	79.79	68.86
Conv-Memnet	78.26	68.38
CNN-ASP	77.82	65.11
TNet	80.79	71.27
DSMN-I	**81.16**	71.50
DSMN-II	**81.16**	**72.09**

modeling, DSMN-II can better utilize the information of the nearby aspects to improve the performance of the samples which contain multiple aspects.

To conduct a more detailed analysis, we sample some qualitative cases randomly from TNet [3], DSMN-I [12] and DSMN-II, as shown in Table IV. Some useful information can be obtained from the table.

Firstly, the introduction of inter-aspect modeling can improve the performance of the multi-aspect cases which are difficult to predict sentiment polarities based on the context. Among the given multi-aspect cases, TNet, which does not consider inter-aspect modeling, only performs well on example 1 since the context words, "great" and "free" explicitly express the sentiments. However, for some other examples like example 2 and 3, where the polarities of some aspects cannot be directly predicted by their complex contexts, TNet performs worse than the DSMN framework, where the analysis of different aspects in the same sentence is beneficial to each other, which is shown in Fig. 5. Taking example 2 as an example, the first aspect "Windows 7" can be predicted correctly by the context word "crashed," while the context of the second aspect "Windows 8" is more complex. Therefore, the analysis on "Windows 7" will be helpful for the classification of "Windows 8."

Secondly, we also can observe that the introduction of semantic dependency information to the Laptop dataset, which contains more grammatical sentences, is effective. As for the first two aspects of example 5, TNet [3] cannot predict the sentiment result of the aspect "USB3" correctly as it considers similar contexts, while DSMN-I and DSMN-II can capture the comparison relation between the two aspects based on the dependency parsing tree.

Moreover, the major improvement of DSMN-II, namely employing aspect-aware context moment learning instead of sentence-aware context moment, can also make the model perform better. In example 3 and 5, DSMN-II can well balance the local information (i.e. the context around the desired aspect) and the global information (i.e. the auxiliary information of other aspects which are close to the desired aspect). However, DSMN-II may perform worse in some multi-aspect cases, as the inter-aspect information is less considered if the aspects are too far from each other. Taking example 6 as the example, the aspect "size" is far from another aspect "display," so the information of the aspect "display" is less used while the aspect "display" is classified.

Finally, none of the three models can solve some difficult problems in aspect-based sentiment classification. Take example 4 as the example, the three models cannot identify that the context "half an hour" expresses a negative sentiment to the aspects and then predict the sentiment polarities to neutral polarity wrongly.

D. Performance on Restaurant Dataset

We also examine the effectiveness of our model on another dataset about products, namely, the Restaurant dataset.

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TABLE VI
CASE STUDY ON RESTAURANT

Examples	TNet	DSMN-I	DSMN-II
1. However, the **[service]**$_-$ is absolutely horrible.	[service]$_-$ (✓)	[service]$_-$ (✓)	[service]$_-$ (✓)
2. And the **[food]**$_+$, well the **[food]**$_+$ will keep you coming back.	[food]$_+$ (✓), [food]$_+$ (✓)	[food]$_+$ (✓), [food]$_+$ (✓)	[food]$_+$ (✓), [food]$_+$ (✓)
3. The **[fajita]**$_-$ we tried was tasteless and burned and the **[mole sause]**$_-$ was way too sweet.	[fajita]$_-$ (✓), [mole sause]$_+$ (✗)	[fajita]$_-$ (✓), [mole sause]$_-$ (✓)	[fajita]$_-$ (✓), [mole sause]$_-$ (✓)
4. Had **[dinner]**$_0$ here on a Friday and the **[food]**$_+$ was great.	[dinner]$_0$ (✓), [food]$_+$ (✓)	[dinner]$_+$ (✗), [food]$_+$ (✓)	[dinner]$_0$ (✓), [food]$_+$ (✓)

TABLE VII
EXPERIMENTAL RESULTS OF COMPARISON WITH BASELINES ON TWITTER (%)

Models	ACC	Macro-F1
TD-LSTM	66.62	64.01
MemNet	62.50	66.91
BILSTM-ATT-G	72.70	70.84
RAM	71.88	70.33
Conv-Memnet	72.11	70.80
CNN-ASP	73.27	71.77
TNet	**74.97**	**73.60**
DSMN-I	72.25	70.54
DSMN-II	73.84	72.12

The results of performance comparison are shown in Table V. As is shown in Table V, DSMN achieves better performance than all the other methods. Compared to DSMN-I, DSMN-II achieves about 0.6% higher Macro-F1 result. In combination with the performance of DSMN-II on the Laptop dataset, we can conclude that applying aspect-level context moment instead of sentence-level context moment can better capture both the semantic and relation information of the nearby aspects.

Table VI shows some cases sampled randomly from TNet [3], DSMN-I [12] and DSMN-II. From the table, we observe that all of the three models can perform well in some simple cases. However, DSMN-II performs the best in some complicated multi-aspect cases compared to TNet [3] and DSMN-I [12]. Take example 3 as the example, TNet [3] ignores the information of "fajita" while analyzing the sentiment of the aspect "mole sause," so it predicts the sentiment of "mole sause" wrongly according to the context "too sweet". However, the proposed models which exploit the inter-aspect information, namely DSMN-II and DSMN-I, can predict the results of example 3 correctly. In addition, as shown by example 4, DSMN-II gets the correct results of all aspects but DSMN-I has one wrong result. The reason is that the inter-aspect relation learning module of DSMN-II considers the importance of other aspects' information based on the distance between the desired aspect and other aspects.

E. Performance on Twitter Dataset

Since the aspect-based sentiment classification problem exists on some ungrammatical datasets like tweets as well, we also do some experiments on the Twitter dataset and the experimental results are shown in Table VII. The experimental results show that DSMN can achieve better performance than all the baselines except TNet. Compared to DSMN-I, which applies semantic-based information on all datasets, DSMN-II employs location-based distance instead of semantic-based information to guide the multi-hop attention of the memory network. Therefore, DSMN-II improves the performance significantly compared to DSMN-I as DSMN-II avoids the interference of possibly wrong semantic information of ungrammatical sentences. In addition, we can observe that some CNN-based methods (e.g., CNN-ASP and TNet) can achieve good performance on the Twitter dataset, which indicates that CNN-based methods may be more suitable for ungrammatical datasets. However, the proposed DSMN-II still has 0.57% and 0.35% improvements than CNN-ASP in ACC and Macro-F1.

TNet outperforms our framework DSMN in the Twitter dataset, demonstrating the effectiveness of CNN on ungrammatical datasets like the Twitter dataset. Our proposed DSMN does not adopt the CNN operation, which may weaken the performance of our proposed method.

Table VIII gives some qualitative cases randomly from TNet [3], DSMN-I [12] and DSMN-II. From the table, we can see that all three models perform well in some simple cases. However, DSMN-I [12] performs worse in some cases compared to TNet [3] and DSMN-II [12] as it may introduce wrong prior semantic dependency information to the model. With the location-based selection, DSMN-II can well address the above issue and achieve comparable results to TNet [3].

F. Performance on All Datasets

Based on the experimental results on all three datasets, we can see that the DSMN framework performs more effectively than all of the previous methods. Specifically, DSMN achieves better performance than all of the previous methods on Laptop and Restaurant datasets by at least 1.05% ACC, 2.16% F1, and 0.37% ACC, 0.81% F1, respectively. Besides, DSMN performs better than previous methods expect TNet by at least 0.57% ACC, 0.35% F1 on the Twitter dataset.

To further examine the effectiveness of our proposed framework, we conduct experiments on re-sampled datasets by executing ten times, whose results are shown in Table IX. We adopt the t-test and use the p-value p to evaluate the significance level. As the results show, DSMN performs significantly better than the previous state-of-the-art method TNet ($p < 0.01$) by 4.10% and 2.60% F1 on Laptop and Restaurant datasets, respectively, while TNet sightly outperform our model by only 0.47% F1 ($p > 0.05$), and $p > 0.05$ means that TNet doesn't have the significant improvement compared to DSMN. Besides, our method also outperforms RAM significantly ($p < 0.01$), which is also an effective method for aspect level sentiment classification Compared to DSMN-I, DSMN-II has significant improvements ($p < 0.01$) on all three datasets by 1.28%, 1.64% and 2.82% F1.

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TABLE VIII
CASE STUDY ON TWITTER

Examples	TNet	DSMN-I	DSMN-II
1. **[Britney Spears]**$_0$ trouble ...	[Britney Spears]$_0$ (✓)	[Britney Spears]$_0$ (✓)	[Britney Spears]$_0$ (✓)
2. Woods ensures victory for Americans: **[Tiger Woods]**$_0$ delivered the clinching point on Sunday as the United States.	[Tiger Woods]$_0$ (✓)	[Tiger Woods]$_0$ (✓)	[Tiger Woods]$_0$ (✓)
3. Simply_Brianah _ Hayley_G Europe ... Don't ask how I know .. LoL. RT LaLaLaLove_Jas : Where's **[Lady Gaga]**$_0$ ..	[Lady Gaga]$_-$ (✗)	[Lady Gaga]$_0$ (✓)	[Lady Gaga]$_0$ (✓)
4. I'm just waiting for **[Jimmy Carter]**$_0$ to say that all those who think Obama doesn't deserve the nobel are racists.	[Jimmy Center]$_0$ (✓)	[Jimmy Center]$_-$ (✗)	[Jimmy Center]$_0$ (✓)
5. Musicmonday best **[Britney Spears]**$_0$ cover ... ever . oops I did it again ..	[Britney Spears]$_0$ (✓)	[Britney Spears]$_+$ (✗)	[Britney Spears]$_0$ (✓)
6. If Justin Bieber, **[Demi Lovato]**$_+$ or Cody Simpson don't follow you. Don't worry, I help us -LRB- MUST following TheBieberBear !!!	[Demi Lovato]$_+$ (✓)	[Demi Lovato]$_-$ (✗)	[Demi Lovato]$_-$ (✗)
7. I think I'd get most of the way through the fire but burn, losing it forever. Then I'd settle for going back for the **[wii]**$_+$.	[wii]$_0$ (✗)	[wii]$_0$ (✗)	[wii]$_0$ (✗)

TABLE IX
EXPERIMENTAL RESULTS OF RE-SAMPLED DATASETS (%)

Models	Laptop		Restaurant		Twitter	
	ACC	Macro-F1	ACC	Macro-F1	ACC	Macro-F1
RAM	68.70 ± 0.67	65.41 ± 0.80	73.28 ± 0.76	63.23 ± 0.73	66.61 ± 0.97	65.50 ± 0.65
TNet	69.94 ± 1.03	66.63 ± 1.10	73.83 ± 0.44	65.50 ± 0.59	**72.69 ± 1.09**	**70.90 ± 1.18**
DSMN-I	72.12 ± 0.93	69.45 ± 0.93	75.28 ± 0.59	66.46 ± 0.62	68.76 ± 0.82	67.61 ± 0.98
DSMN-II	**73.48 ± 0.30**	**70.73 ± 0.98**	**76.49 ± 0.81**	**68.10 ± 0.34**	71.97 ± 0.86	70.43 ± 1.15

TABLE X
EXPERIMENTAL RESULTS OF ABLATION STUDY (%). M_{sa}, M_{im} AND M_{ir} STAND FOR MULTI-HOP SELECTIVE ATTENTION MODULE, WEIGHTING INTER ASPECT MODELING MODULE AND INTER-ASPECT RELATION LEARNING MODULE, RESPECTIVELY

Models	Laptop		Restaurant	
	Acc.	F1	Acc.	F1
M_{sa}	75.21	71.87	79.95	70.53
M_{sa}+M_{im}	76.90	72.93	81.00	71.49
M_{sa}+M_{ir}	76.30	72.03	80.05	70.76
M_{sa}+M_{im}+M_{ir}	**77.27**	**73.91**	**81.16**	**72.09**

V. EXTENSIVE EXPERIMENTS

A. Ablation Study

To examine the effectiveness of each module of the DSMN framework, we conduct the ablation study based on the DSMN-II model. The experimental results are shown in Table X. From the experimental results, we observe that DSMN with only multi-hop selective attention can outperform memory-network-based methods, like MemNet and RAM, which shows the effectiveness of the multi-hop selective attention module. Besides, applying the weighting inter-aspect modeling and inter-aspect relation learning can also improve the performance.

Moreover, we give some qualitative cases on the effectiveness of inter-aspect modules, shown in Table XI. Taking the sentence "The fajita we tried was tasteless and burned and the mole sause was way too sweet." as an example, the DSMN framework can correctly predict the sentiment polarities of both the aspect "fajita" and "mole sause". However, when we remove the inter-aspect modules, the model can only predict the aspect "fajita" correctly, and the aspect "mole sause" cannot be predicted correctly since the model cannot identify the sentiment of too sweet and exploit the information of the other aspect "fajita" to assist the classification of "mole sause."

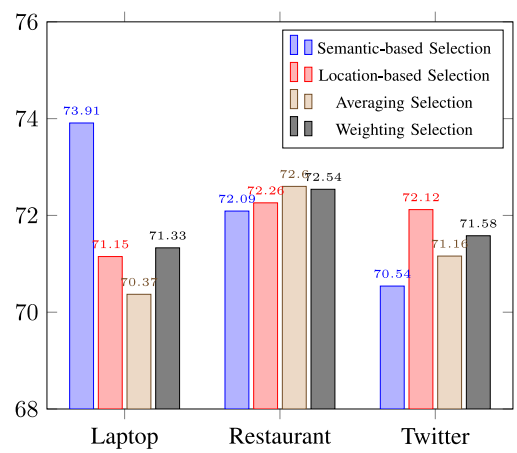

Fig. 6. Macro F1 measures (%) of comparison between different selection strategies.

B. Effects of Different Selection Strategies

To verify that different selection strategies have different performances on different types of datasets. We conduct extensive experiments on the three datasets, and the results are shown in Fig. 6. From the results, we can easily observe that different selection strategies have different performances on the three datasets. Semantic-based distance has a better performance on Laptop reviews, which confirms that semantic dependency information can improve the performance for some datasets which contain more formal sentences. However, semantic dependency information may introduce some noises to the model for some other datasets containing ungrammatical sentences like Twitter. Therefore, location-based distance can have a better performance in these datasets.

Moreover, among the four strategies, the weighting selection achieves the second-best performance on the three datasets.

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TABLE XI
CASES W/O INTER-ASPECT MODULES

Examples	DSMN without Inter-Aspect Modules	DSMN with Inter-Aspect Modules
1. Great **[price]**$_+$ free **[shipping]**$_+$ what else can I ask for ! !	[price]$_+$ (✓), [shipping]$_+$ (✓)	[price]$_+$ (✓), [shipping]$_+$ (✓)
2. My laptop with **[Windows 7]**$_-$ crashed and I did not want **[Windows 8]**$_-$.	[Windows 7]$_-$ (✓), [Windows 8]$_0$ (✗)	[Windows 7]$_-$ (✓), [Windows 8]$_-$ (✓)
3. The **[fajita]**$_-$ we tried was tasteless and burned and the **[mole sause]**$_-$ was way too sweet.	[fajita]$_-$ (✓), [mole sause]$_+$ (✗)	[fajita]$_-$ (✓), [mole sause]$_-$ (✓)

TABLE XII
EXPERIMENTAL RESULTS OF COMPARISON BETWEEN DIFFERENT ASPECT MODELING METHODS (%)

Models	Laptop		Restaurant		Twitter	
	ACC	Macro-F1	ACC	Macro-F1	ACC	Macro-F1
Embedding+Mean Pooling	74.45	69.77	80.18	70.59	73.12	71.23
Embedding+Max Pooling	75.39	70.09	80.00	70.87	72.83	71.46
Embedding+Attention	74.61	70.16	80.09	70.70	72.83	71.08
BiLSTM+Mean Pooling	76.18	71.68	80.54	71.47	73.12	71.55
BiLSTM+Max Pooling	75.39	71.25	80.89	71.79	72.54	71.07
BiLSTM+Attention	**77.27**	**73.91**	**81.16**	**72.09**	**73.84**	**72.12**

Therefore, we can adopt the weighting selection for any given dataset without specifying the adopted distance, since it can learn the probability of adopting semantic-based distance for each sample and achieve comparable results. Compared to the weighting selection, the averaging selection performs worse, since it just simply treats the two kinds of distance equally.

As for the Restaurant dataset, all of the four selection strategies are effective for the Restaurant dataset as they have similar performances on the Restaurant dataset. After analyzing both Restaurant and Laptop datasets, we can observe that the Restaurant dataset is less complex than the Laptop dataset. Specifically, the sentences with the length less than 20 words account for more in Restaurant than Laptop (73.2% vs. 68.6%), and the aspects with only a single word also account for more in Restaurant (73.5% vs. 59.2%). All the above observations can show that the Restaurant dataset is the relatively easier dataset to solve compared to the Laptop dataset, even if they have the same genre, namely review. Therefore, as for the Restaurant dataset, some simple settings like adopting location-based distance can also capture effective information and achieve competitive results compared to some complex settings like introducing semantic-based distance.

C. Effects of Aspect Modeling Methods

To verify whether the aspect encoding method which consists of a BiLSTM layer and an attention layer can better model aspects for improving the performance, we conduct some extensive experiments. The experimental results are shown in Table XII. As the table shows, we can see that both Laptop and Restaurant perform better with the BiLSTM layer which can capture the relationship between words of aspects. However, applying the BiLSTM network does not have a significant improvement of the performance on Twitter since the aspects of the Twitter dataset are mostly names, which usually have no obvious structure. In addition, applying the attention mechanism is also effective as it can capture the important part of the words of aspects.

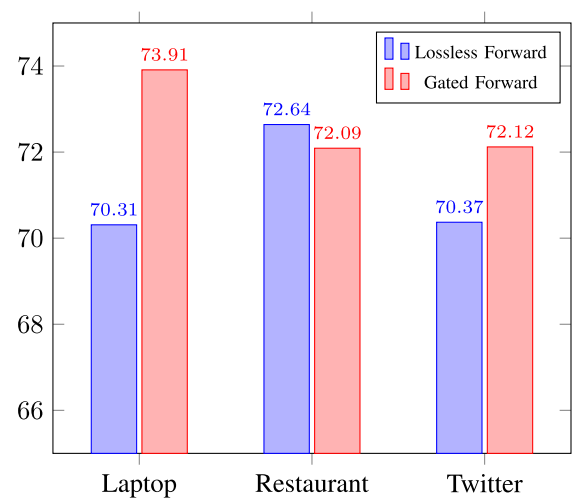

Fig. 7. Macro F1 measures (%) of comparison between different attention combination methods.

D. Effects of Multi-Hop Attention Combination Methods

To examine whether gated forward performs better than lossless forward for combining the attention results for the multi-hop memory network, we conduct extensive experiments and the results are shown in Fig. 7. As the table shows, we can observe that gated forward can improve the performance of the model well compared to lossless forward on both Laptop and Twitter datasets. Therefore, it is obvious that the model with gated forward can better capture the important attention results for the sentiment classification task since gated forward can control how much the outputs of the current layer that is useful for identifying the sentiment polarity of the desired aspect.

VI. CONCLUSION

In this paper, we have designed a novel framework, namely Deep Selective Memory Network (DSMN) for aspect level sentiment classification. Based on the DSMN framework, we design

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the basic model DSMN-I and the advanced model DSMN-II. In the proposed DSMN-I, semantic dependency information is introduced to the deep memory network and a sentence-level context moment task is designed to learn the relation among aspects to better classify the aspects. As for DSMN-II, we provide different types of information, namely semantic-based information and location-based information, to select the context memory dynamically for the multi-hop memory network. In addition, DSMN-II integrates the information of the nearby aspects for better predicting the sentiment of the desired aspect by designing an aspect-aware context moment learning task and a weighting inter-aspect modeling module. Experiments on three benchmark datasets show that our framework can perform well in various datasets. The extensive experimental results also show the effectiveness of our framework.

Since integrating pre-trained language models and knowledge bases to natural language processing tasks are prevalent these years, it is necessary to do some research and exploration with these ideas in the future. Moreover, extracting aspects and then predicting the sentiment polarities of the extracted aspects is also a hot topic recently, we will also try to make some exploration in this interesting task.

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